Atanu Kundu

Senior Research Fellow,
Indian Association for the Cultivation of Science, Kolkata, India.

Gmail Twitter LinkedIn GitLab
Google scholar atanucs.github.io

Research Area

- I am a researcher in formal verification and falsification of cyber-physical systems, with expertise in developing algorithms for bounded model checking (BMC) and machine learning–based falsification. Key contributions include:

  - Developed a CEGAR-based BMC algorithm for affine hybrid systems, which integrates SAT-solving based path enumeration and path-wise reachability analysis, enabling efficient verification of safety properties.
  - Designed a bounded reachability algorithm for compositional affine hybrid systems, specifically tailored for compositional systems, thereby improving scalability in the analysis of large-scale CPS.
  - Designed surrogate model–based falsification frameworks for industrial CPS, bridging data-driven techniques with robustness monitoring to address the limitations of traditional optimization-based falsification methods. In particular, deep neural networks and decision trees are employed as surrogate models of the system under test.
  - Developed a causation-aware falsification strategy leveraging reinforcement learning and causation monitoring to efficiently accelerate counterexample searching in CPS.

Education

2020 – present	**PhD in Computer Science from IACS Kolkata, India**. **Thesis title:** *Bounded Model Checking and Data-Driven Falsification strategies for CPS.* **Courses:** Big Data Analytics and Advanced Machine Learning **CGPA:** 8.50
2016 – 2018	**M.Sc. in Computer Science from Visva-Bharati, India**. **Courses:** Advanced Data Structure, AI, DAA, etc **CGPA:** 7.93
2013 – 2016	**B.Sc. in Computer Science from The University of Burdwan**. **Percentage:** 66.12

Research Publications

1. S. Gon, A. Kundu, and R. Ray, "Falsification of cyber-physical systems using causation-aware reinforcement learning," *accepted in 29th ACM International Conference on Hybrid Systems: Computation and Control, HSCC/ICCPS,*, 2026.

2. L. Bu, A. Kundu, R. Ray, and Y. Shi, "Arch-comp25 category report: Hybrid systems with piecewise constant dynamics and bounded model checking," in *Proceedings of 12th Int. Workshop on Applied Verification for Continuous and Hybrid Systems*, G. Frehse and M. Althoff, Eds., ser. EPiC Series in Computing, vol. 108, EasyChair, 2025, pp. 1–14. DOI: 10.29007/x3nv.

3. T. Khandait et al., "Arch-comp25 category report: Falsification," in *Proceedings of 12th Int. Workshop on Applied Verification for Continuous and Hybrid Systems*, G. Frehse and M. Althoff, Eds., ser. EPiC Series in Computing, vol. 108, EasyChair, 2025, pp. 169–189. DOI: 10.29007/dgnn.

4. A. Kundu, S. Gon, and R. Ray, "Data-driven falsification of cyber-physical systems," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, 2025. DOI: 10.1109/TCAD.2025.3608632.
5. L. Bu, A. Kundu, R. Ray, and Y. Shi, "Arch-comp24 category report: Hybrid systems with piecewise constant dynamics and bounded model checking," in *Proceedings of the 11th Int. Workshop on Applied Verification for Continuous and Hybrid Systems*, G. Frehse and M. Althoff, Eds., ser. EPiC Series in Computing, vol. 103, EasyChair, 2024, pp. 1–14. DOI: 10.29007/nv67.
6. T. Khandait et al., "Arch-comp 2024 category report: Falsification," in *Proceedings of the 11th Int. Workshop on Applied Verification for Continuous and Hybrid Systems*, G. Frehse and M. Althoff, Eds., ser. EPiC Series in Computing, vol. 103, EasyChair, 2024, pp. 122–144. DOI: 10.29007/hgfv.
7. A. Kundu, S. Gon, and R. Ray, "Data-driven falsification of cyber-physical systems," in *Proceedings of the 17th Innovations in Software Engineering Conference*, ser. ISEC '24, , Bangalore, India, Association for Computing Machinery, 2024, ISBN: 9798400717673. DOI: 10.1145/3641399.3641401.
8. A. Kundu, S. Das, and R. Ray, "Sat-reach: A bounded model checker for affine hybrid systems," *ACM Trans. Embed. Comput. Syst.*, vol. 22, no. 2, Jan. 2023, ISSN: 1539-9087. DOI: 10.1145/3567425.
9. C. Menghi et al., "Arch-comp23 category report: Falsification," in *Proceedings of 10th International Workshop on Applied Verification of Continuous and Hybrid Systems (ARCH23)*, G. Frehse and M. Althoff, Eds., ser. EPiC Series in Computing, vol. 96, EasyChair, 2023, pp. 151–169. DOI: 10.29007/6nqs.
10. L. Bu, G. Frehse, A. Kundu, R. Ray, Y. Shi, and E. Zaffanella, "Arch-comp22 category report: Hybrid systems with piecewise constant dynamics and bounded model checking," in *Proceedings of 9th International Workshop on Applied Verification of Continuous and Hybrid Systems (ARCH22)*, vol. 90, 2022, pp. 44–57. DOI: 10.29007/lnzf.

Under Review

- A. Kundu, P. Sarkar, R. Ray, "A CEGAR-centric Bounded Reachability Analysis for Compositional Affine Hybrid Systems ". [Journal]

Projects and Work Experiences

2017 - May 2018 — Ralay TCP over wireless network with free-space optical and worldwide interoperability for microwave access.
under the supervision of Mr. Subhasis Banerjee, Visva-Bharati, India.

May - Dec 2022 — Building neural network models of the standard Cyber-Physical Systems (CPS) benchmarks.
Under the supervision of Dr. Rajarshi Ray, IACS Kolkata, India.
Building Feed-forward Neural Networks (FNNs) for CPS benchmarks such as hybrid automata and Simulink models. We built FNN models for systems with both linear and non-linear dynamics, as well as time-varying input signals. These models are approximations of the original models and can be useful in developing applications for such systems.

Skills

- **Languages** — Native language Bengali. Fluent in English and Hindi.
- **Coding** — Proficient in C, C++, Python, and SMT-LIB2. Familiar with MATLAB and LaTeX.
- **Technical Domains** — Formal Verification, Hybrid Automata, Falsification, Reachability Analysis, SAT Solving, STL, Robustness Monitoring, and Surrogate Modeling using Machine Learning.

Skills (continued)

DevOps	Experience in git and Docker.
Tools	FlexiFal, NNFal, XSpeed, SAT-Reach, SpaceEx, Flow*, dReach, and breach.
Soft Skill	Team Work, Time Management, Presentation Skills, and Leadership.
Misc.	Academic research, teaching, training, consultation, LaTeX typesetting, and publishing.

Teaching Experiences

Spring 2022 — Teaching assistant on Artificial Intelligence (COM 4211)
Responsibilities included setting and evaluating lab assignments. Supervising students throughout the course to succeed in the lab assignments.

Autumn 2023 — Teaching assistance on Object-Oriented Programming with C++ Lab (COM 4111)
My duties involved creating and assessing assignments and surprise tests. Additionally, I was responsible for overseeing students' progress during the course, ensuring their success in laboratory assignments.

Miscellaneous Experience

Selected Talks

- 2025 — Research paper **Data-driven Falsification of Cyber-physical Systems** accepted at IEEE TCAD 2025 and presented at RHPL co-located with FSTTCS 2025, BITS Pilani, GOA.
- 2024 — Research paper **Data-driven Falsification of Cyber-physical Systems** accepted at ISEC 2024 and presented the paper at IIIT Bangalore.
- 2023 — A short presentation titled **A Framework for Detecting Unsafe Behavior in Cyber-Physical Systems** presented in Tech Symposium on Computing Trends 2023, organized by THALES associated with University of Calcutta.
- 2022 — **SAT-Reach: A Bounded Model Checker for Affine Hybrid Systems** presented in the formal methods update meeting 2022 at IIT Delhi.

Certification

- 2023 — **Special recognition for completing a short thesis presentation in exactly 5 minutes**. Awarded by THALES associated with University of Calcutta.

References

Dr. Rajarshi Ray
Associate Professor
School of Mathematical & Computational Sciences,
Indian Association for the Cultivation of Science,
Kolkata, India
rajarshi.ray@iacs.res.in.

Dr. Ansuman Banerjee
Professor
Advanced Computing and Microelectronics Unit,
Indian Statistical Institute, Kolkata, India
ansuman@isical.ac.in

Dr. Partha Basuchowdhuri
Associate Professor
School of Mathematical & Computational Sciences,
Indian Association for the Cultivation of Science,
Kolkata, India
partha.basuchowdhuri@iacs.res.in.